

Unit IX: Electronic Devices

10 Periods

Chapter–14: Semiconductor Electronics: Materials, Devices and Simple Circuits

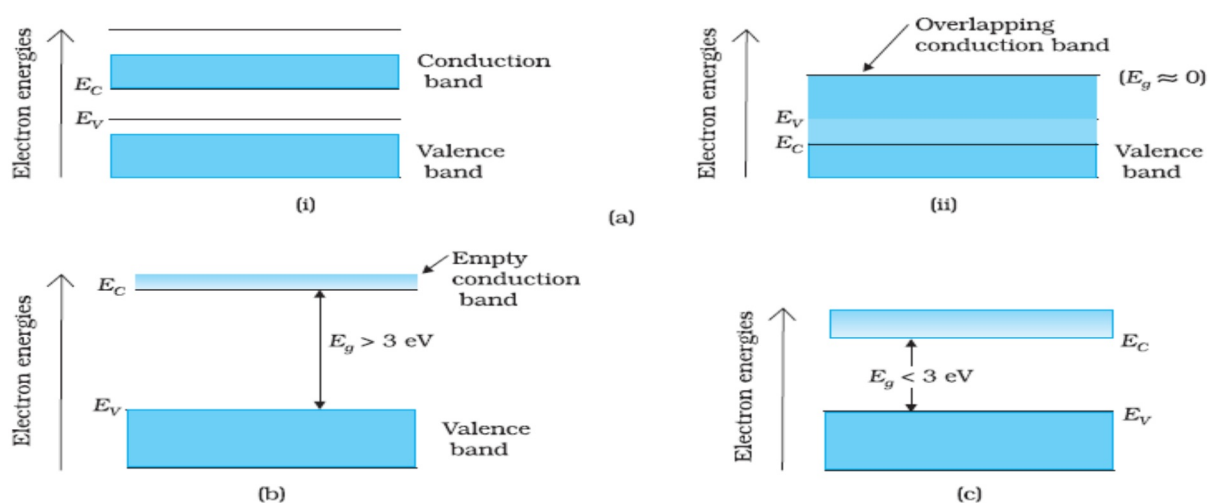
GIST OF THE CHAPTER:

Energy bands in conductors, semiconductors and insulators (qualitative ideas only) Intrinsic and extrinsic semiconductors- p and n type, p-n junction Semiconductor diode - I-V characteristics in forward and reverse bias, application of junction diode -diode as a rectifier.

Energy bands in solids:

- Due to influence of high electric field between the core of the atoms and the shared electrons, energy levels are split-up or spread-out forming energy bands.
- The energy band formed by a series of levels containing valance electrons is called valance band and the lowest unfilled energy level just above the valance band is called conduction band.
- Filled energy levels are separated from the band of unfilled energy levels by an energy gap called **forbidden gap** or energy gap or band gap.

Energy band diagram for, Conductors Semiconductors and Insulators



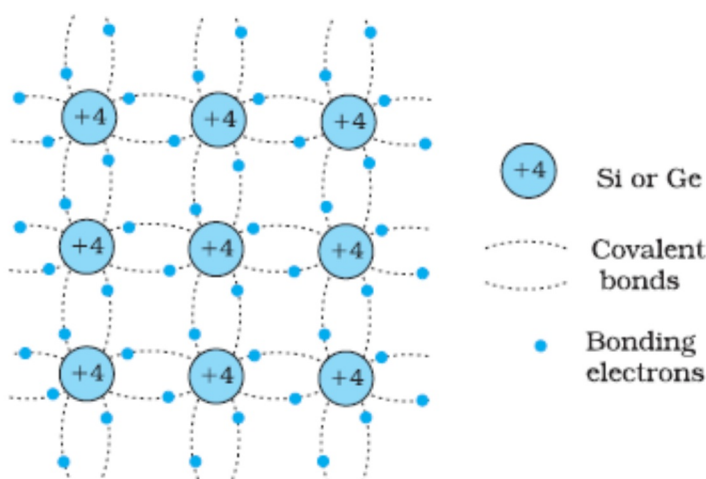
Conductors (Metals): The conduction band and valance band partly overlap each other and there is no forbidden energy gap in between. Large number of electrons are available for electrical conduction, hence the resistance is low of such materials. Even if a small electric field is applied across the metal, these free electrons start moving. Hence metals behave as a conductor.

Semiconductors: The conduction and valance bands are separated by the small energy gap (1 eV) called forbidden energy gap. The valance band is completely filled, while the conduction band is empty at zero kelvin. The electrons cross from valance band to conduction band even when a small amount of energy is supplied. The semiconductor acquires a small conductivity at room temperature.

Insulators: Electrons, however heated, cannot practically jump to conduction band from valence band due to a large energy gap (>3 eV). Therefore, conduction is not possible in insulators.

INTRINSIC SEMICONDUCTORS:

- Intrinsic Semiconductor is a pure semiconductor.
- The energy gap in Si is 1.1 eV and in Ge is 0.74 eV.
- Si: $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$. (Atomic No. is 14)
- Ge: $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^2, 4p^2$. (Atomic No. is 32)
- Both Si and Ge have four valance electrons. The four valance electrons form four covalent bonds by sharing the electrons with neighbouring four atoms.
- In intrinsic semiconductor, the number of thermally generated electrons always equals the number of holes. So, if n_i and p_i are the concentration of electrons and holes respectively, then $n_i = p_i$. The quantity n_i or p_i is referred to as the 'intrinsic carrier concentration'.
- At 0 K, a semiconductor is an insulator i.e., it possesses zero conductivity. When temperature is increased, a few covalent bonds break up and release the electrons. These electrons move to conduction band leaving behind equal number of holes in valence band. The conductivity of an intrinsic semiconductor is due to both electrons and holes.



DOPING:

- Doping a Semiconductor: Doping is the process of deliberate addition of a very small amount of impurity (1% of crystal atoms) into an intrinsic semiconductor.
- The impurity atoms are called dopants.
- The semiconductor containing impurity is known as '**Extrinsic semiconductor**'.
- Doping of a semiconductor increases its electrical conductivity to a great extent.
- The pentavalent impurity atoms are called donor atoms, while the trivalent impurity atoms are called acceptor atoms

Extrinsic semiconductor.

A semiconductor doped with a suitable impurity (pentavalent or trivalent), so as to possess conductivity much higher than the semiconductor in pure form is called an extrinsic semiconductor.

Extrinsic semiconductors are of two types:

- 1) n-type semiconductor
- 2) p-type semiconductor

1) n-type semiconductor:

When a pentavalent impurity, such as arsenic or antimony or phosphorus is added to a pure semiconductor, the four of the five valance electrons of the impurity atoms form covalent bonds by sharing the electrons with the adjoining four silicon atoms, while the fifth electron is very loosely bound with the parent impurity atom and is comparatively free to move.

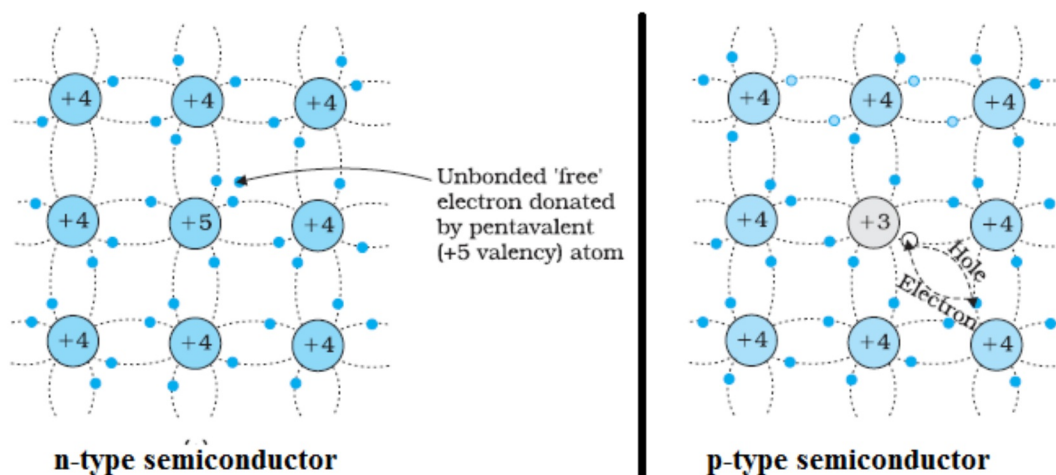
The number of free electrons become more than the holes in the semiconductor and such an extrinsic semiconductor is called n-type semiconductor. In other words, in a n-type semiconductor, electrons are majority carriers and holes are minority carriers.

2) p-type semiconductor:

When a trivalent impurity, such as indium or gallium or boron is added to a pure semiconductor, three valance electrons of the impurity atoms form covalent bonds by sharing the electrons with the adjoining three silicon atoms.

Due to the deficiency of an electron, there is one incomplete covalent bond. The vacancy that exists with the fourth covalent bond with fourth Si atom constitutes a hole.

The semiconductor becomes deficient in electrons i.e. number of holes become more than the number of electrons. Such a semiconductor is called p-type semiconductor. It has holes as majority carriers and electrons as minority carriers.



Electrical conductivity of a semiconductor:

The conductivity of a semiconductor is determined by the mobility (μ) of both electrons and holes and their concentration. Mathematically- $\sigma = e (n_e \mu_e + n_h \mu_h)$

P-N JUNCTION.

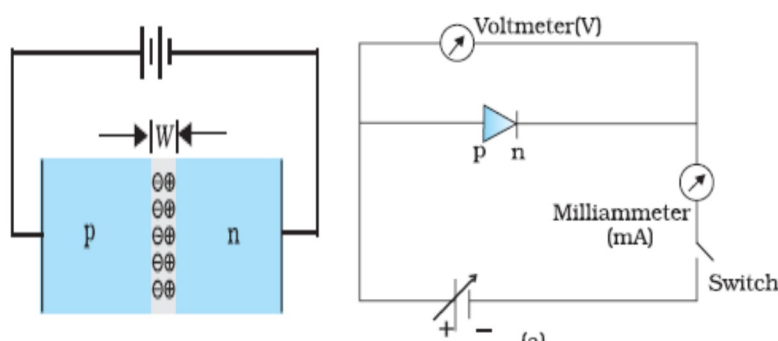
The device obtained by bringing a p-type semiconductor crystal into close contact with n-type semiconductor crystal is called a p-n junction. It conducts in one direction only. It is also called a junction diode

Depletion layer. It is a thin layer formed between the p and n-sections and devoid of holes and electrons. Its width is about 10^{-8} m. A potential difference of about 0.7 V is produced across the junction, which gives rise to a very high electric field ($= 10^6$ V/ m).

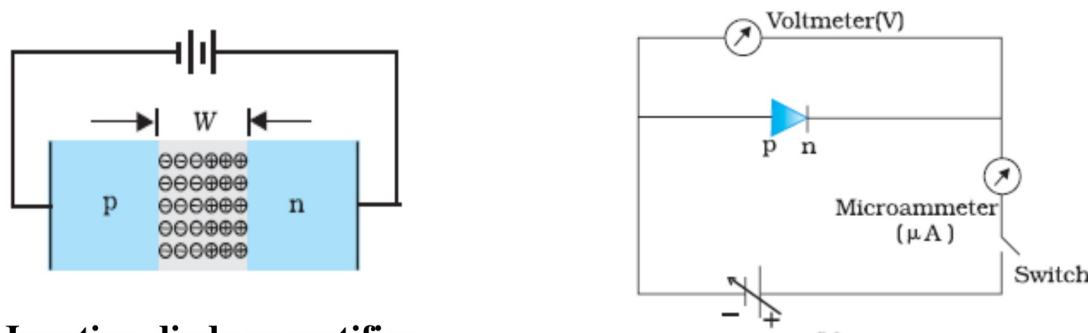
Potential Barrier: The difference in potential between p and n regions across the junction makes it difficult for the holes and electrons to move across the junction. This acts as a barrier and hence called ‘potential barrier’. **Potential barrier for Si is nearly 0.7 V and for Ge is 0.3 V. The potential barrier opposes the motion of the majority carriers.**

Forward biasing:

The p-n junction is said to be forward biased, when the positive terminal of the external battery is connected to p-section and the negative terminal to n-section of the junction diode.



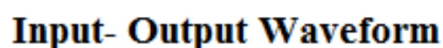
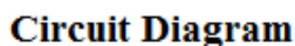
Reverse biasing: The p-n junction is said to be reverse biased, when the positive terminal of the battery is connected to n-section and the negative terminal to p-section of the junction diode.



Junction diode as rectifier:

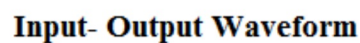
Because of its unidirectional conduction property, the p-n junction is used to convert an a.c. voltage into d. c. voltage, It is, then, said to be acting as a rectifier.

1. **Half wave rectifier:** A rectifier, which rectifies only one half of each a.c. input supply cycle, is called a half wave rectifier. A half wave rectifier gives discontinuous and pulsating d.c. output. As alternative half cycles of the a.c. input supply go waste, its efficiency is very low.



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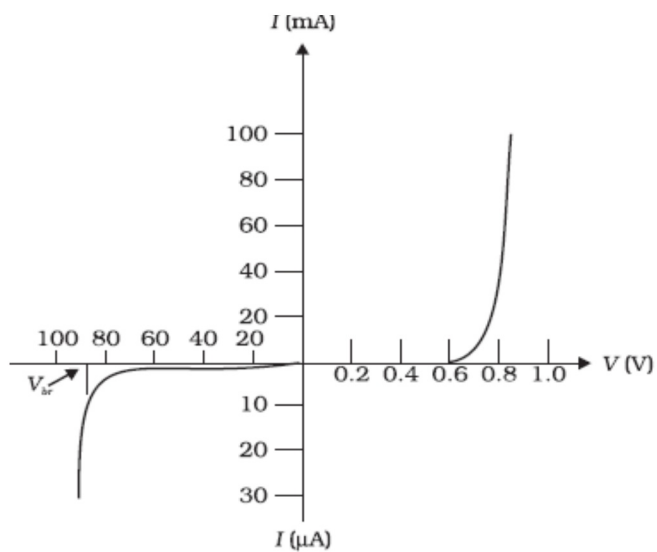
Circuit Diagram



GRAPHS

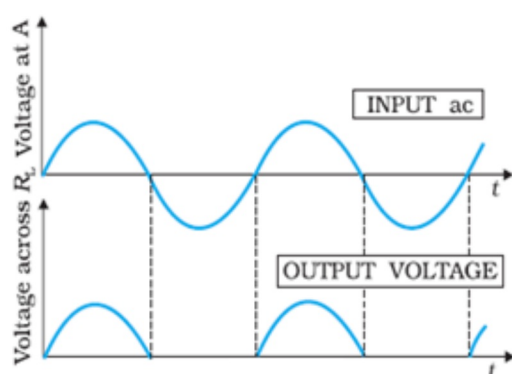
1) I -V CHARACTERISTICS:

Forward Bias & Reverse Bias Characteristics of a P-N Junction Diode

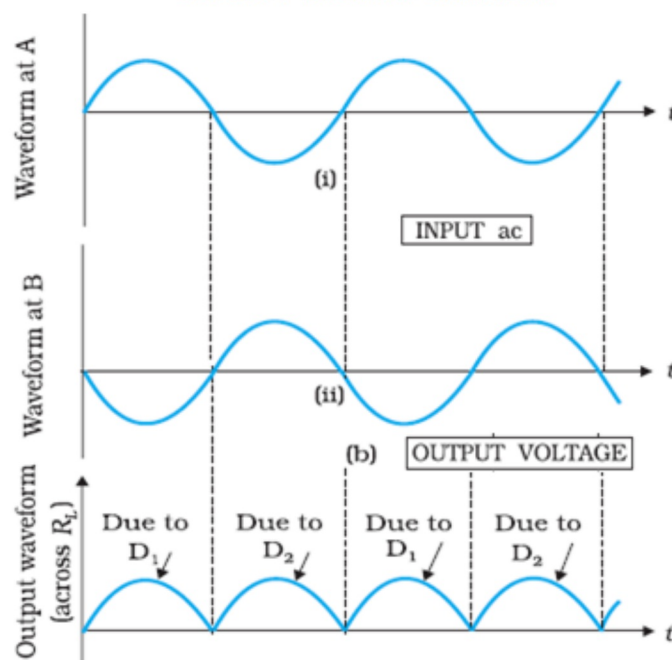


2) INPUT AND OUTPUT VOLTAGE GRAPHS OF

A HALF WAVE RECTIFIER

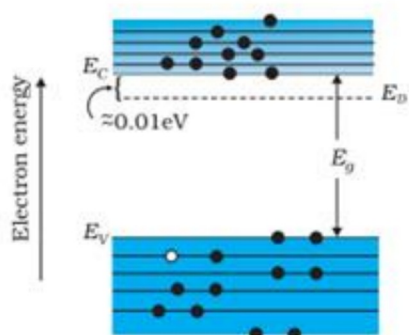


A FULL WAVE RECTIFIER

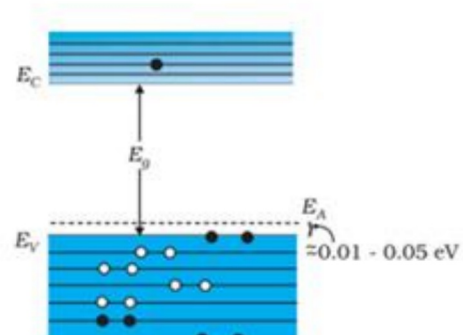


ENERGY BAND DIAGRAMS IN EXTRINSIC SEMICONDUCTORS

1) n-TYPE SEMICONDUCTOR

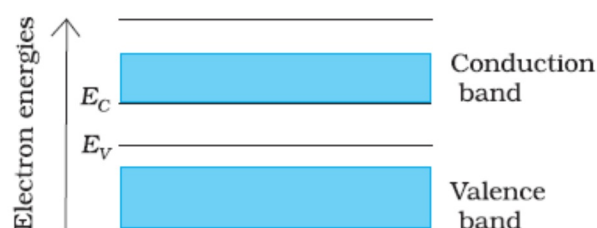


2) p-TYPE SEMICONDUCTOR

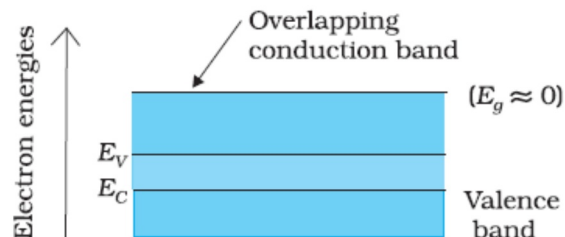


ENERGY BAND DIAGRAMS IN CONDUCTORS, INSULATORS AND SEMICONDUCTOS

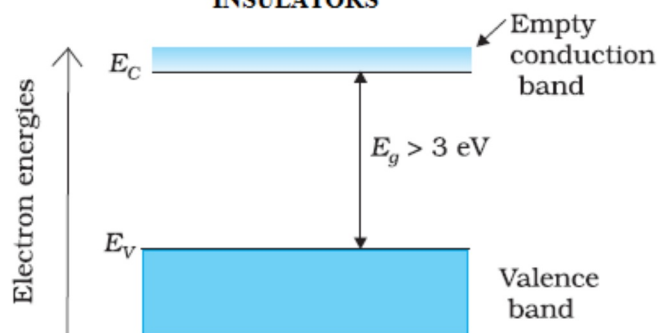
CONDUCTORS HAVING PARTIALLY FILLED CONDUCTION BAND



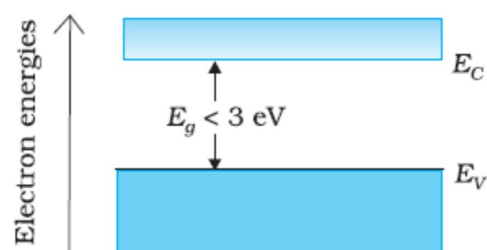
CONDUCTORS WITH OVERLAPPING CONDUCTION AND VALENCE BAND



INSULATORS



SEMICONDUCTORS



TABLES

1) DIFFERENCE BETWEEN INTRINSIC AND EXTRINSIC SEMICONDUCTORS

S.NO	INTRINSIC SEMICONDUCTOR	EXTRINSIC SEMICONDUCTOR
1	Pure form of semiconductor.	Impure form of semiconductor.
2	Conductivity is low	Conductivity is higher than intrinsic semiconductor.
3	The no of holes is equal to no of free electrons	In n-type, the no. of electrons is greater than that of the holes and in p-type, the no. holes is greater than that of the electrons.
4	The conduction depends on temperature.	The conduction depends on the concentration of doped impurity and temperature.

2) DIFFERENCE BETWEEN HALF WAVE AND FULL WAVE RECTIFIER

S.NO	HALF WAVE RECTIFIER	FULL WAVE RECTIFIER
1	Only half cycle of AC is rectified.	Both cycles of AC are rectified.
2	Requires only one diode	Requires two diodes.
3	The output frequency is equal to input supply frequency. (F)	The output frequency is double of the input supply frequency. (2F)
4	The electric current through the load is not continuous	A continuous electric current flow through the load.

FORMULAE

1) Electron and hole concentration in a semiconductor in thermal equilibrium

$$n_e n_h = n_i^2$$

2) Resistance of a Diode:

- a) Static or DC Resistance $R_{dc} = V/I$
- b) Dynamic or AC Resistance

$$R_{a.c} = \Delta V / \Delta I$$

MNEMONICS

1) TO REMEMBER NAMES OF IMPURITIES IN SEMICONDUCTORS

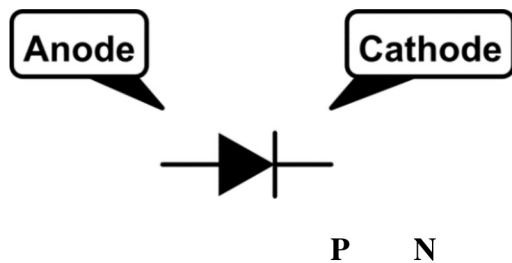
BIG PAA

Boron, Indium, Gallium (all three are trivalent impurities)

Phosphorus, Antimony, Arsenic (all three are pentavalent impurities)

2) TO REMEMBER THE P AND N SECTIONS OF A DIODE.

The arrow in the schematic symbol for diodes points in the direction of Conventional (positive) current flow.



3) Current is **unidirectional** in a diode. It flows from anode to cathode only. To remember this, remember the mnemonics ‘ACID’ (ANODE CURRENT IN DIODE)